

1. An application encrypts data using an AWS KMS customer master key (CMK) with imported key material. The CMK is referenced by an alias in the application code. Company policy mandates that the CMK must be rotated every 6 months. What is the process to rotate the key?

- A. Import new key material into a new CMK, update the key alias to point to the new CMK.
- B. Use an AWS managed CMK with automatic rotation every 6 months. Update the alias.
- C. Delete the current key material and import new material into the existing CMK.
- D. Enable automatic key rotation for the CMK and specify a period of 6 months.

Answer A

2. (Select Two) A company is in the process of creating a web application on Amazon Web Services (AWS) and is utilizing Amazon CloudFront with a `www.sample.com` domain name. Ensuring encrypted traffic in transit to CloudFront is a crucial requirement, and an SSL certificate for `www.sample.com` has been created through AWS Certificate Manager (ACM).

What two procedures must a CloudOps Engineer execute to guarantee that the traffic is encrypted in transit?

- A. Add the alternate domain name (CNAME) of `www.sample.com` into the CloudFront distribution and select the custom SSL certificate.
- B. Establish CloudFront Origin Shield for the original CloudFront source.
- C. Adjust the Viewer Protocol Policy in every cache behavior in the CloudFront distribution to reroute HTTP traffic to HTTPS.
- D. Set up an AWS WAF web ACL associated with the CloudFront distribution.
- E. Change the Viewer Protocol Policy for each cache behavior in the CloudFront distribution to permit both HTTP and HTTPS traffic.

Answer A and C

3. A company has recently rolled out an application in a production environment. The environment currently operates on a single Amazon EC2 instance which hosts both the application's web interface and a MySQL database. As per company policy, all production IT environments need to maintain high availability. What action should a CloudOps Engineer undertake to comply with this requirement?

- A. Transition the database from the EC2 instance to an Amazon RDS for MySQL Multi-AZ DB instance. Use the AWS Application Migration Service to refactor the application into an AWS Lambda function. Deploy the Lambda function using the Multi-AZ option.
- B. Transition the database from the EC2 instance to an Amazon RDS for MySQL Multi-AZ DB instance. Deploy the application on EC2 instances within an Auto Scaling group distributed across multiple Availability Zones. Configure a load balancer in front of the EC2 instances.
- C. Replicate the database onto another EC2 instance in a different Availability Zone. Use AWS Backup to create Amazon Machine Images (AMIs) of both the application EC2 instance and the database EC2 instance. Develop an AWS Lambda function that conducts health checks every minute. In the event of a failure, program the Lambda function to initiate a new EC2 instance from the AMIs created by AWS Backup.
- D. Transfer the database to a different EC2 instance. Situate the application EC2 instance in an Auto Scaling group spanning multiple Availability Zones. Produce an Amazon Machine Image (AMI) from the database EC2 instance. Use this AMI to launch a secondary database EC2 instance in a different Availability Zone. Keep the second database EC2 instance in a stopped state. Employ the second database EC2 instance as a standby.

Answer B

4. A company manages a fleet of Amazon EC2 instances in a VPC and wishes to remove their public IP addresses to protect them from internet-based threats. Some applications still require access to Amazon S3 buckets. A CloudOps Engineer has been tasked with providing continued access to the S3 buckets. Which solutions can the Administrator recommend?

- A. Add an outbound rule in the security groups of the EC2 instances for Amazon S3 using private IP addresses.
- B. Configure the internet gateway to route connections to S3 using private IP addresses.
- C. Create a VPC endpoint in the VPC and configure the route tables appropriately.
- D. Deploy a NAT gateway in a public subnet and configure the route tables in the VPC appropriately.
- E. Set up AWS Direct Connect and configure a virtual interface between the EC2 instances and the S3 buckets.

Answer C and D

5. An application is being deployed that hosts highly sensitive data that must not be leaked outside of the company. A security team has asked a CloudOps Engineer to configure the environment to address this concern. How can the Administrator ensure that the servers in the VPC cannot send traffic to the internet?

- A. Ensure that the servers do not have Elastic IP addresses.
- B. Create a blackhole NAT gateway that prevents outbound access.
- C. Launch the EC2 instances in private subnets.
- D. Use instance stores to ensure there is no persistent data.

Answer C

6. (Select Two) A company is creating a new application that will run in a hybrid environment. The application processes data that must be secured and the developers require encryption in-transit across shared networks and encryption at rest. Which combination of actions should a CloudOps Engineer take to meet these requirements?

- A. Use AWS KMS to create TLS/SSL certificates.
- B. Configure an AWS VPN between the on-premises data center and AWS.
- C. Use AWS Certificate Manager to create TLS/SSL certificates.
- D. Use AWS CloudHSM to encrypt the data using a CMK.
- E. Use AWS KMS to manage the encryption keys used for data encryption.

Answer B and E

7. (Select Two) An application records highly sensitive customer data to several Amazon S3 buckets. The S3 buckets are secured with bucket policies. There have been reports of attempts at unauthorized access and the security team have requested that information about which buckets are being targeted and by whom is gathered. Which steps should a CloudOps Engineer gather the requested information?

- A. Use Amazon Athena to query the S3 Server Access Logs for HTTP 503 errors and determine the IAM user or role making the requests.
- B. Configure Amazon S3 Analytics on all of the affected S3 buckets and generate a report showing the unauthorized access attempts.
- C. Configure Amazon S3 Server Access Logging on all of the affected S3 buckets and store the logs in a separate, dedicated bucket.
- D. Use Amazon Athena to query S3 Analytics reports for HTTP 403 errors and determine the IAM user or role making the requests.
- E. Use Amazon Athena to query the S3 Server Access Logs for HTTP 403 errors, and determine the IAM user or role making the requests.

Answer C and E

8. A security consultant has identified unnecessary security group and network ACL rules that pose a security risk. What steps should a CloudOps Engineer take to resolve the security vulnerabilities?

- A. Contact AWS Support and notify them of the vulnerabilities.
- B. Use Amazon Inspector to identify security best practices and rectify issues.
- C. Create an AWS WAF web ACL that protects resources from web attacks.
- D. Remove the unnecessary security group rules and network ACL rules.

Answer D

9. A CloudOps Engineer has stored the login credentials for a database as secure string parameters in AWS Systems Manager Parameter Store. An application running on an Amazon EC2 instance must use the credentials to access the database. What is the MOST secure way to grant the application access to the credentials?

- A. Create an IAM role for the EC2 instances and grant the role permission to read the Systems Manager parameters.
- B. Create an IAM group for the application and grant the group permission to read the Systems Manager parameters.
- C. Create an IAM policy for the application and grant the policy permission to read the Systems Manager parameters.
- D. Create an IAM user for the application and grant the user permission to read the Systems Manager parameters.

Answer A

10. Each IT staff member in a company uses a unique IAM user account. Permissions are applied to users using IAM policies and IAM groups. The security team has requested that staff members should log in with their on-premises Active Directory user accounts instead of their IAM user accounts when accessing the AWS Management Console. Which solution can a CloudOps Engineer implement to the requirements of the security team?

- A. Enable an Active Directory federation in an Amazon Route 53 private zone.
- B. Implement a two-way trust relationship between AWS IAM and Active Directory.
- C. Implement a VPN tunnel and configure an Active Directory connector.
- D. Use the IAM connector to synchronize the on-premises Active Directory.

Answer C

11. A company operates a web-based application that is deployed on an Amazon EC2 instance. The users have been reporting delays during peak business hours. An examination of the Amazon CloudWatch metrics for the application by a CloudOps Engineer shows that the CPU usage of the instance frequently exceeds 90% during these peak periods. Which is the MOST operationally efficient solution to increase the efficiency of the application?

- A. Configure an Auto Scaling group that is tied to an Application Load Balancer. Configure a target tracking scaling policy triggered by the average CPU utilization of the Auto Scaling group.
- B. Activate Amazon CloudWatch logs on the EC2 instance and establish a CloudWatch alarm for CPU usage that alerts the CloudOps Engineer whenever the CPU usage exceeds 90%.
- C. Set up an AWS Site-to-Site VPN connection allowing application users to directly link to the private IP address of the EC2 instance, thus minimizing latency.
- D. Create a CloudWatch alarm that gets triggered when the CPU usage of the EC2 instance goes beyond 80%. Configure this alarm to initiate an AWS Lambda function that performs vertical scaling of the instance.

Answer A

12. A CloudOps Engineer plans to use a single AWS CloudFormation template to create and manage stacks across multiple AWS accounts and regions with a single operation. What feature of AWS CloudFormation will help the Administrator to accomplish this?

- A. Change sets
- B. Nested stacks
- C. StackSets
- D. Stack policies

Answer C

13. A company's CloudOps Engineer deploys four new Amazon EC2 instances by using the standard Amazon Linux 2 Amazon Machine Image (AMI). The company needs to be able to use AWS Systems Manager to manage the instances. The CloudOps Engineer notices that the instances do not appear in the Systems Manager console. What must the CloudOps Engineer do to resolve this issue?

- A. Connect to each instance by using SSH. Install Systems Manager Agent on each instance. Configure Systems Manager Agent to start automatically when the instances start up.
- B. Use AWS Certificate Manager (ACM) to create a TLS certificate. Import the certificate into each instance. Configure Systems Manager Agent to use the TLS certificate for secure communications.
- C. Connect to each instance by using SSH. Create an ssm-user account. Add the ssm-user account to the /etc/sudoers directory.
- D. Attach an IAM instance profile to the instances. Ensure that the instance profile contains the AmazonSSMManagedInstanceCore policy.

Answer D

13. A corporation runs an application exclusively on Amazon EC2 Spot Instances. These instances operate within an EC2 Auto Scaling group with scheduled scaling actions configured. The operations team have noted that the capacity doesn't consistently scale at the scheduled intervals, and instances face numerous terminations within a single day. It's the responsibility of a CloudOps Engineer to ensure timely instance launch with fewer disruptions. Which solution would satisfy these requirements?

- A. Use the capacity-optimized allocation strategy for Spot Instances and broaden the range of instance types within the Auto Scaling group.
- B. Use the capacity-optimized allocation strategy for Spot Instances and augment the maximum size of the Auto Scaling group.
- C. Use the on-demand instance allocation strategy and expand the range of instance types within the Auto Scaling group.
- D. Use the on-demand instance allocation strategy and increase the maximum size of the Auto Scaling group.

Answer A

14. A company's CloudOps Engineer makes routine checks of the AWS Health Dashboard across all the company's accounts. These accounts are part of an organization within AWS Organizations. The company recently integrated 10 more accounts into the organization. The CloudOps Engineer needs to aggregate the alerts from the Health Dashboard of each account. What is the most efficient approach to meet this requirement?

- A. Configure AWS CloudWatch Events in each account to capture events from the Health Dashboard and forward them to a centralized AWS CloudWatch Logs.
- B. Construct an AWS Lambda function to interrogate the AWS Health API and record all events into an Amazon Aurora database.
- C. Activate organizational view in AWS Health.
- D. Employ the AWS Health API to record events into an Amazon S3 bucket.

Answer C

15. A CloudOps Engineer noticed a large increase in the number of requests against an Amazon SQS queue. The administrator is concerned about rising costs associated with the queue and needs to identify the source of the calls. What should the CloudOps Engineer use to validate the calls made to SQS?

- A. Amazon SQS Access Logs
- B. Amazon CloudWatch
- C. AWS Cost Explorer
- D. AWS CloudTrail

Answer D

16. (Select Two) A CloudOps Engineer manages an application running on an Auto Scaling group of Amazon EC2 instances behind an Application Load Balancer (ALB). The database layer uses Amazon RDS MySQL with an Amazon ElastiCache for Memcached cluster as a caching layer. The CloudOps Engineer must perform all system patching at 10pm on a Wednesday. Which resources require the configuration of a maintenance window?

- A. Amazon EC2 instances
- B. Amazon ElastiCache cluster
- C. Amazon RDS instance
- D. Amazon EC2 Auto Scaling
- E. Elastic Load Balancer

Answer B and C

17. An application server running on an Amazon EC2 instance recently failed due to an Amazon EBS volume running out of space. The failure caused an outage of a critical application. Which steps should a CloudOps Engineer take to prevent this from happening again?

- A. Create an AWS Lambda function that monitors the disk space metrics using the Amazon EBS API. Post a notification to an Amazon SNS topic when disk space is running low.
- B. Enable detailed monitoring for the EC2 instances. Create an Amazon CloudWatch alarm to notify the Administrator when disk space is running low.
- C. Install the Amazon CloudWatch agent on the EC2 instance to collect disk metrics. Create a CloudWatch alarm to notify the Administrator when disk space is running low.
- D. Configure Amazon CloudWatch Events to monitor Amazon EC2 status checks for the status of the EBS volumes. Post a notification to an Amazon SNS topic to notify the if the disk is impaired.

Answer C

18. A company uses AWS Service Catalog to manage approved services. A new AWS account has been created and a CloudOps Engineer needs to create a replica of the company's existing AWS infrastructure in the new AWS account. Currently, an AWS Service Catalog portfolio is used to create and manage resources. What is the MOST efficient way to accomplish this?

- A. Share the AWS Service Catalog portfolio with the other AWS account and import the portfolio into the AWS account.
- B. Manually create an AWS Service Catalog portfolio in the new AWS account and recreate the original portfolio.
- C. Create an AWS CloudFormation template to redeploy the AWS Service Catalog portfolio in the new AWS account.
- D. Run an AWS Lambda function to create a new AWS Service Catalog portfolio based on the output of the DescribePortfolio API operation.

Answer A

19. A company's website has been running slowly during busy periods. The website runs on Amazon EC2 instances and uses an Amazon RDS database. The CloudOps Engineer suspects the issue is related to high CPU usage on a component of this application. How should the CloudOps Engineer investigate which component is causing the performance bottleneck?

- A. Use Amazon CloudWatch metrics to examine the resource usage of each component.
- B. Use Amazon Inspector to view the detailed resource usage for each component.
- C. Use Amazon CloudWatch Logs and Amazon Athena to search for utilization data.
- D. Use AWS CloudTrail to review the API usage metrics for each component.

Answer A

20. An environment consisting of 100 EC2 Windows instances. The Amazon CloudWatch agent is deployed and running across all the instances with a baseline configuration file to capture log files. There is a new requirement to capture the DHCP log files that exist on 50 of the instances. What is the most operationally efficient way to meet this?

- A. Create an additional CloudWatch agent configuration to capture the DHCP logs.
- B. Create a Powershell Script to push the baseline configuration
- C. Run the CloudWatch agent configuration on each EC2 instance and verify the baseline config
- D. Run the CloudWatch agent configuration on each EC2 instance and select the advanced detail level

Answer A

21. A company is partnering with an external vendor to provide data processing services. For this integration, the vendor must host the company's data in an Amazon S3 bucket in the vendor's AWS account. The vendor is allowing the company to provide an AWS Key Management Service (AWS KMS) key to encrypt the company's data. The vendor has provided an IAM role Amazon Resource Name (ARN) to the company for this integration. What should a CloudOps Engineer do to configure this integration?

- A. Create a new KMS key. Add the vendor's IAM role ARN to the KMS key policy. Provide the new KMS key ARN to the vendor.
- B. Create a new KMS key. Create a new IAM user. Add the vendor's IAM role ARN to an inline policy that is attached to the IAM user. Provide the new IAM user ARN to the vendor.
- C. Configure encryption using the KMS managed S3 key. Add the vendor's IAM role ARN to the KMS managed S3 key policy. Provide the KMS managed S3 key ARN to the vendor.
- D. Configure encryption using the KMS managed S3 key. Create a S3 bucket. Add the vendor's IAM role ARN to the S3 bucket policy. Provide the S3 bucket ARN to the vendor.

Answer A

22. A company is managing many accounts by using a single organization in AWS Organizations. The organization has all features enabled. The company wants to turn on AWS Config in all the accounts of the organization and in all AWS Regions. What should a CloudOps Engineer do to meet these requirements in the MOST operationally efficient way?

- A. Use AWS CloudFormation StackSets to deploy stack instances that turn on AWS Config in all accounts and in all Regions.
- B. Use AWS CloudFormation StackSets to deploy stack policies that turn on AWS Config in all accounts and in all Regions.
- C. Use Service Control Policies (SCPs) to configure AWS Config in all accounts and in all Regions.
- D. Create a script that uses the AWS CLI to turn on AWS Config in all accounts in the organization. Run the script from the organization's management account.

Answer A

23. A CloudOps Engineer needs to give users the ability to upload objects to an Amazon S3 bucket. The CloudOps Engineer creates a presigned URL and provides the URL to a user, but the user cannot upload an object to the S3 bucket. The presigned URL has not expired, and no bucket policy is applied to the S3 bucket. Which of the following could be the cause of this problem?

- A. The user has not properly configured the AWS CLI with their access key and secret access key.
- B. The CloudOps Engineer does not have the necessary permissions to upload the object to the S3 bucket.
- C. The CloudOps Engineer must apply a bucket policy to the S3 bucket to allow the user to upload the object.
- D. The object already has been uploaded through the use of the presigned URL, so the presigned URL is no longer valid.

Answer B

24. A company migrated an I/O intensive application to an Amazon EC2 general purpose instance. The EC2 instance has a single General Purpose SSD Amazon Elastic Block Store (Amazon EBS) volume attached. Application users report that certain actions that require intensive reading and writing to the disk are taking much longer than normal or are failing completely. After reviewing the performance metrics of the EBS volume, a CloudOps Engineer notices that the VolumeQueueLength metric is consistently high during the same times in which the users are reporting issues. The CloudOps Engineer needs to resolve this problem to restore full performance to the application. Which action will meet these requirements?

- A. Modify the instance type to be storage optimized.
- B. Modify the volume properties by deselecting Auto-Enable Volume 10.
- C. Modify the volume properties to increase the IOPS.
- D. Modify the instance to enable enhanced networking.

Answer C

25. A company creates custom AMI images by launching new Amazon EC2 instances from an AWS CloudFormation template it installs and configure necessary software through AWS OpsWorks and takes images of each EC2 instance. The process of installing and configuring software can take between 2 to 3 hours but at times the process stalls due to installation errors. The CloudOps Engineer must modify the CloudFormation template so if the process stalls, the entire stack will fail and roll back. Based on these requirements what should be added to the template?

- A. Conditions with a timeout set to 4 hours.
- B. CreationPolicy with timeout set to 4 hours.
- C. DependsOn a timeout set to 4 hours.
- D. Metadata with a timeout set to 4 hours.

Answer B

26. A company asks a CloudOps Engineer to ensure that AWS CloudTrail files are not tampered with after they are created. Currently, the company uses AWS Identity and Access Management (IAM) to restrict access to specific trails. The company's security team needs the ability to trace the integrity of each file. What is the MOST operationally efficient solution that meets these requirements?

- A. Create an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function when a new file is delivered. Configure the Lambda function to compute an MD5 hash check on the file and store the result in an Amazon DynamoDB table. The security team can use the values that are stored in DynamoDB to verify the integrity of the delivered files.
- B. Create an AWS Lambda function that is invoked each time a new file is delivered to the CloudTrail bucket. Configure the Lambda function to compute an MD5 hash check on the file and store the result as a tag in an Amazon S3 object. The security team can use the information in the tag to verify the integrity of the delivered files.
- C. Enable the CloudTrail file integrity feature on an Amazon S3 bucket. Create an IAM policy that grants the security team access to the file integrity logs that are stored in the S3 bucket.
- D. Enable the CloudTrail file integrity feature on the trail. The security team can use the digest file that is created by CloudTrail to verify the integrity of the delivered files

Answer D

27. A company plans to run a public web application on Amazon EC2 instances behind an Elastic Load Balancer (ELB). The company's security team wants to protect the website by using AWS Certificate Manager (ACM) certificates. The ELB must automatically redirect any HTTP requests to HTTPS. Which solution will meet these requirements?

- A. Create an Application Load Balancer that has one HTTPS listener on port 80. Attach an SSL/TLS certificate to listener port 80. Create a rule to redirect requests from HTTP to HTTPS.
- B. Create an Application Load Balancer that has one HTTP listener on port 80 and one HTTPS protocol listener on port 443. Attach an SSL/TLS certificate to listener port 443. Create a rule to redirect requests from port 80 to port 443.
- C. Create an Application Load Balancer that has two TCP listeners on port 80 and port 443. Attach an SSL/TLS certificate to listener port 443. Create a rule to redirect requests from port 80 to port 443.
- D. Create a Network Load Balancer that has two TCP listeners on port 80 and port 443. Attach an SSL/TLS certificate to listener port 443. Create a rule to redirect requests from port 80 to port 443.

Answer C

28. A company wants to use only IPv6 for all its Amazon EC2 instances. The EC2 instances must not be accessible from the internet, but the EC2 instances must be able to access the internet. The company creates a dual-stack VPC and IPv6-only subnets. How should a CloudOps Engineer configure the VPC to meet these requirements?

- A. Create and attach a NAT gateway. Create a custom route table that includes an entry to point all IPv6 traffic to the NAT gateway. Attach the custom route table to the IPv6-only subnets.
- B. Create and attach an internet gateway. Create a custom route table that includes an entry to point all IPv6 traffic to the internet gateway. Attach the custom route table to the IPv6-only subnets.
- C. Create and attach an egress-only internet gateway. Create a custom route table that includes an entry to point all IPv6 traffic to the egress-only internet gateway. Attach the custom route table to the IPv6-only subnets.
- D. Create and attach an internet gateway and a NAT gateway. Create a custom route table that includes an entry to point all IPv6 traffic to the internet gateway and all IPv4 traffic to the NAT gateway. Attach the custom route table to the IPv6-only subnets.

Answer C

29. A company uses Amazon Elasticsearch Service (Amazon ES) to analyze sales and customer usage data. Members of the company's geographically dispersed sales team are traveling. They need to log in to Kibana by using their existing corporate credentials that are stored in Active Directory. The company has deployed Active Directory Federation Services (AD FS) to enable authentication to cloud services. Which solution will meet these requirements?

- A. Configure Active Directory as an authentication provider in Amazon ES. Add the Active Directory server's domain name to Amazon ES. Configure Kibana to use Amazon ES authentication.
- B. Deploy an Amazon Cognito user pool. Configure Active Directory as an external identity provider for the user pool. Enable Amazon Cognito authentication for Kibana on Amazon ES.
- C. Enable Active Directory user authentication in Kibana. Create an IP-based custom domain access policy in Amazon ES that includes the Active Directory server's IP address.
- D. Establish a trust relationship with Kibana on the Active Directory server. Enable Active Directory user authentication in Kibana. Add the Active Directory server's IP address to Kibana.

Answer B

30. A CloudOps Engineer is responsible for a legacy CPU-heavy application. The application can only be scaled vertically. Currently, the application is deployed on a single t2 large Amazon EC2 instance. The system is showing 90% CPU usage and significant performance latency after a few minutes. What change should be made to alleviate the performance problem?

- A. Change the Amazon EBS volume to Provisioned IOPS.
- B. Upgrade to a compute-optimized instance.
- C. Add additional t2.large instances to the application.
- D. Purchase Reserved Instances.

Answer B

31. A company recently migrated its application to a VPC on AWS. An AWS Site-to-Site VPN connection connects the company's on-premises network to the VPC. The application retrieves customer data from another system that resides on premises. The application uses an on-premises DNS server to resolve domain records. After the migration, the application is not able to connect to the customer data because of name resolution errors. Which solution will give the application the ability to resolve the internal domain names?

- A. Launch EC2 instances in the VPC. On the EC2 instances, deploy a custom DNS forwarder that forwards all DNS requests to the on-premises DNS server. Create an Amazon Route 53 private hosted zone that uses the EC2 instances for name servers.
- B. Create an Amazon Route 53 Resolver outbound endpoint. Configure the outbound endpoint to forward DNS queries against the on-premises domain to the on-premises DNS server.
- C. Set up two AWS Direct Connect connections between the AWS environment and the on-premises network. Set up a link aggregation group (LAG) that includes the two connections. Change the VPC resolver address to point to the on-premises DNS server.
- D. Create an Amazon Route 53 public hosted zone for the on-premises domain. Configure the network ACLs to forward DNS requests against the on-premises domain to the Route 53 public hosted zone.

Answer B

32. A company runs workloads on 90 Amazon EC2 instances in the eu-west-1 Region in an AWS account. In 2 months, the company will migrate the workloads from eu-west-1 to the eu-west-3 Region. The company needs to reduce the cost of the EC2 instances. The company is willing to make a 1-year commitment that will begin next week. The company must choose an EC2 Instance purchasing option that will provide discounts for the 90 EC2 Instances regardless of Region during the 1-year period. Which solution will meet these requirements?

- A. Purchase EC2 Standard Reserved Instances.
- B. Purchase an EC2 Instance Savings Plan.
- C. Purchase EC2 Convertible Reserved Instances.
- D. Purchase a Compute Savings Plan.

Answer D

33. A company uses AWS Organizations. A CloudOps Engineer wants to use AWS Compute Optimizer and AWS tag policies in the management account to govern all member accounts in the billing family. The CloudOps Engineer navigates to the AWS Organizations console but cannot activate tag policies through the management account. What could be the reason for this issue?

- A. All features have not been enabled in the organization.
- B. Consolidated billing has not been enabled.
- C. The member accounts do not have tags enabled for cost allocation.
- D. The member accounts have not manually enabled trusted access for Compute Optimizer.

Answer A

34. A large company is using AWS Organizations to manage its multi-account AWS environment. According to company policy, all users should have read-level access to a particular Amazon S3 bucket in a central account. The S3 bucket data should not be available outside the organization. A CloudOps Engineer must set up the permissions and add a bucket policy to the S3 bucket. Which parameters should be specified to accomplish this in the MOST efficient manner?

- A. Specify "*" as the principal and PrincipalOrgId as a condition.
- B. Specify all account numbers as the principal.
- C. Specify PrincipalOrgId as the principal.
- D. Specify the organization's management account as the principal

Answer A

35. A company has a stateless application that is hosted on a fleet of 10 Amazon EC2 On-Demand Instances in an Auto Scaling group. A minimum of 6 instances are needed to meet service requirements. Which action will maintain uptime for the application MOST cost-effectively?

- A. Use a Spot Fleet with an On-Demand capacity of 6 instances.
- B. Update the Auto Scaling group with a minimum of 6 On-Demand Instances and a maximum of 10 On-Demand Instances.
- C. Update the Auto Scaling group with a minimum of 1 On-Demand Instance and a maximum of 6 On-Demand Instances.
- D. Use a Spot Fleet with a target capacity of 6 instances.

Answer A

36. (Select Two) An Amazon S3 Inventory report reveals that more than 1 million objects in a S3 bucket are not encrypted. These objects must be encrypted, and all future objects must be encrypted at the time they are written. Which combination of actions should a CloudOps Engineer take to meet these requirements?

- A. Create an AWS Config rule that runs evaluations against configuration changes to the S3 bucket. When an unencrypted object is found run an AWS Systems Manager Automation document to encrypt the object in place.
- B. Edit the properties of the S3 bucket to enable default server-side encryption.
- C. Filter the S3 Inventory report by using S3 Select to find all objects that are not encrypted. Create a S3 Batch Operations job to copy each object in place with encryption enabled.
- D. Filter the S3 Inventory report by using S3 Select to find all objects that are not encrypted. Send each object name as a message to an Amazon Simple Queue Service (Amazon SQS) queue. Use the SQS queue to invoke an AWS Lambda function to tag each object with a key of Encryption and a value of SSE-KMS
- E. Use S3 Event Notifications to invoke an AWS Lambda function on all new object-created events for the S3 bucket. Configure the Lambda function to check whether the object is encrypted and to run an AWS Systems Manager Automation document to encrypt the object in place when an unencrypted object is found

Answer B and C

37. A company has an AWS Site-to-Site VPN connection between on-premises resources and resources that are hosted in a VPC. A CloudOps Engineer launches an Amazon EC2 instance that has only a private IP address into a private subnet in the VPC. The EC2 instance runs Microsoft Windows Server. A security group for the EC2 instance has rules that allow inbound traffic from the on-premises network over the VPN connection. The on-premises environment contains a third-party network firewall. Rules in the third-party network firewall allow Remote Desktop Protocol (RDP) traffic to flow between the on-premises users over the VPN connection. The on-premises users are unable to connect to the EC2 instance and receive a timeout error. What should the CloudOps Engineer do to troubleshoot this issue?

- A. Create Amazon CloudWatch logs for the EC2 instance to check for blocked traffic.
- B. Create Amazon CloudWatch logs for the Site-to-Site VPN connection to check for blocked traffic.
- C. Create VPC flow logs for the EC2 instance's elastic network interface to check for rejected traffic.
- D. Instruct users to use EC2 Instance Connect as a connection method.

Answer C

38. A company hosts a web portal on Amazon EC2 instances. The web portal uses an Elastic Load Balancer (ELB) and Amazon Route 53 for its public DNS service. The ELB and the EC2 instances are deployed by way of a single AWS CloudFormation stack in the us-east-1 Region. The web portal must be highly available across multiple Regions. Which configuration will meet these requirements?

- A. Deploy a copy of the stack in the us-west-2 Region. Create a single start of authority (SOA) record in Route 53 that includes the IP address from each ELB. Configure the SOA record with health checks. Use the ELB in us-east-1 as the primary record and the ELB in us-west-2 as the secondary record.
- B. Deploy a copy of the stack in the us-west-2 Region. Create an additional A record in Route 53 that includes the ELB in us-west-2 as an alias target. Configure the A records with a failover routing policy and health checks. Use the ELB in us-east-1 as the primary record and the ELB in us-west-2 as the secondary record.
- C. Deploy a new group of EC2 instances in the us-west-2 Region. Associate the new EC2 instances with the existing ELB, and configure load balancer health checks on all EC2 instances. Configure the ELB to update Route 53 when EC2 instances in us-west-2 fail health checks.
- D. Deploy a new group of EC2 instances in the us-west-2 Region. Configure EC2 health checks on all EC2 instances in each Region. Configure a peering connection between the VPC's. Use the VPC in us-east-1 as the primary record and the VPC in us-west-2 as the secondary record.

Answer B

39. A company is testing Amazon Elasticsearch Service (Amazon ES) as a solution for analyzing system logs from a fleet of Amazon EC2 instances. During the test phase, the domain operates on a single-node cluster. A CloudOps Engineer needs to transition the test domain into a highly available production-grade deployment. Which Amazon ES configuration should the CloudOps Engineer use to meet this requirement?

- A. Use a cluster of four data nodes across two AWS Regions. Deploy four dedicated master nodes in each Region.
- B. Use a cluster of six data nodes across three Availability Zones. Use three dedicated master nodes.
- C. Use a cluster of six data nodes across three Availability Zones. Use six dedicated master nodes.
- D. Use a cluster of eight data nodes across two Availability Zones. Deploy four master nodes in a failover AWS Region.

Answer B

40. (Select Two) A CloudOps Engineer is investigating why a user has been unable to use RDP to connect over the internet from their home computer to a bastion server running on an Amazon EC2 Windows instance. Which of the following are possible causes of this issue?

- A. A network ACL associated with the bastion's subnet is blocking the network traffic.
- B. The instance does not have a private IP address.
- C. The route table associated with the bastion's subnet does not have a route to the internet gateway.
- D. The security group for the instance does not have an inbound rule on port 22.
- E. The security group for the instance does not have an outbound rule on port 3389.

Answer A and C

41. A company is running a flash sale on its website. The website is hosted on burstable performance Amazon EC2 instances in an Auto Scaling group. The Auto Scaling group is configured to launch instances when the CPU utilization is above 70%. A couple of hours into the sale, users report slow load times and error messages for refused connections. A CloudOps Engineer reviews Amazon CloudWatch metrics and notices that the CPU utilization is at 20% across the entire fleet of instances. The CloudOps Engineer must restore the website's functionality without making changes to the network infrastructure. Which solution will meet these requirements?

- A. Activate unlimited mode for the instances in the Auto Scaling group.
- B. Implement an Amazon CloudFront distribution to offload the traffic from the Auto Scaling group.
- C. Move the website to a different AWS Region that is closer to the users.
- D. Reduce the desired size of the Auto Scaling group to artificially increase CPU average utilization.

Answer A

42. A CloudOps Engineer is using AWS Systems Manager Patch Manager to patch a fleet of Amazon EC2 instances. The CloudOps Engineer has configured a patch baseline and a maintenance window. The CloudOps Engineer also has used an instance tag to identify which instances to patch. The CloudOps Engineer must give Systems Manager the ability to access the EC2 instances. Which additional action must the CloudOps Engineer perform to meet this requirement?

- A. Add an inbound rule to the instances' security group.
- B. Attach an IAM instance profile with access to Systems Manager to the instances.
- C. Create a Systems Manager activation Then activate the fleet of instances.
- D. Manually specify the instances to patch Instead of using tag-based selection.

Answer B

43. A CloudOps Engineer has created an AWS Service Catalog portfolio and has shared the portfolio with a second AWS account in the company. The second account is controlled by a different engineer. Which action will the Engineer of the second account be able to perform?

- A. Add a product from the imported portfolio to a local portfolio.
- B. Add new products to the imported portfolio.
- C. Change the launch role for the products contained in the imported portfolio.
- D. Customize the products in the imported portfolio

Answer A

44. (Select Two) A CloudOps Engineer must configure a resilient tier of Amazon EC2 instances for a high performance computing (HPC) application. The HPC application requires minimum latency between nodes. Which actions should the CloudOps Engineer take to meet these requirements?

- A. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the file system to the EC2 instances by using user data.
- B. Create a Multi-AZ Network Load Balancer in front of the EC2 instances.
- C. Place the EC2 instances in an Auto Scaling group within a single subnet.
- D. Launch the EC2 instances into a cluster placement group.
- E. Launch the EC2 instances into a partition placement group

Answer C and D

45. A gaming application is deployed on four Amazon EC2 instances in a default VPC. The CloudOps Engineer has noticed consistently high latency in responses as data is transferred among the four instances. There is no way for the Engineer to alter the application code. The MOST effective way to reduce latency is to relaunch the EC2 instances in:

- A. Dedicated VPC.
- B. Single subnet inside the VPC.
- C. Placement group.
- D. Single Availability Zone

Answer C

46. A CloudOps Engineer is investigating a company's web application for performance problems. The application runs on Amazon EC2 instances that are in an Auto Scaling group. The application receives large traffic increases at random times throughout the day. During periods of rapid traffic increases, the Auto Scaling group is not adding capacity fast enough. As a result, users are experiencing poor performance. The company wants to minimize costs without adversely affecting the user experience when web traffic surges quickly. The company needs a solution that adds more capacity to the Auto Scaling group for larger traffic increases than for smaller traffic increases. How should the CloudOps Engineer configure the Auto Scaling group to meet these requirements?

- A. Create a simple scaling policy with settings to make larger adjustments in capacity when the system is under heavy load.
- B. Create a step scaling policy with settings to make larger adjustments in capacity when the system is under heavy load.
- C. ■ Create a target tracking scaling policy with settings to make larger adjustments in capacity when the system is under heavy load.
- D. Use Amazon EC2 Auto Scaling lifecycle hooks. Adjust the Auto Scaling group's maximum number of instances after every scaling event.

Answer B

47.(Select Two) A CloudOps Engineer has an AWS CloudFormation template created from an infrastructure stack deployed in us-east-1. The Administrator attempts to use the template to launch a stack in us-west-1. The stack partially deploys but then errors and rolls back. What are the most likely reasons for the failure of the stack deployment?

- A. The template referenced services that do not exist in us-west-1.
- B. The template referenced an Amazon Machine Image (AMI) that is not available in us-west-1.
- C. The template did not have the proper level of permissions to deploy the resources in us-west-1.
- D. CloudFormation templates can be used only to deploy stacks in a single Region.
- E. The template referenced an IAM user that is not available in us-west-1.

Answer A & B

48. (Select Two) A university is using a distributed computing application to run complex calculations on a fleet of Amazon EC2 instances. The calculations will be running for at least 1 year. The application uses 2 control nodes and between 8 and 24 worker nodes as required. The control nodes run continuously while the worker nodes run for 8 hours a day and are launched when required. What is the MOST cost-effective pricing model for the application?

- A. Use Reserved Instances for the worker nodes.
- B. Use Spot Instances for the worker nodes and On-Demand Instances if there is no Spot availability.
- C. Use Spot Instances for the control nodes and On-Demand Instances if there is no Spot availability.
- D. Use Dedicated Hosts for the control nodes.
- E. Use Reserved Instances for the control nodes.

Answer B & E

49. A CloudOps Engineer needs to replicate an application setup, currently running on numerous Amazon EC2 instances within the eu-west-1 Region, to three additional AWS Regions. The current setup in eu-west-1 utilizes specifically configured Amazon Machine Images (AMIs) and an AWS CloudFormation template for resource deployment. What is the MOST efficient way for the CloudOps Engineer to replicate this setup across the additional Regions?

- A. Use AWS Elastic Beanstalk to deploy the application in each additional region, leveraging the original application configuration.
- B. Deploy an EC2 instance in each Region, manually configure them as per the original instance, and then create new AMIs. Update the CloudFormation template to utilize the new AMIs.
- C. Modify the CloudFormation template to add the new Regions to the Auto Scaling group, then update the existing stack in eu-west-1.
- D. Use the `aws ec2 copy-image` command to copy the AMI to each new Region. Modify the CloudFormation template to include mappings for the copied AMIs.

Answer D

50. A company security and compliance policy mandates that specific AMIs are approved for usage in the company's AWS accounts. A CloudOps Engineer needs to review existing Amazon EC2 instances to ensure they are in compliance with this policy. Which action should a CloudOps Engineer take?

- A. Create a custom report using AWS Systems Manager Inventory to identify unapproved AMIs.
- B. Create an AWS Config rule to check that approved AMI IDs have been used.
- C. Create an AWS Lambda function that checks that approved AMI IDs have been used.
- D. Use AWS Trusted Advisor to identify EC2 workloads using unapproved AMIs.

Answer B

51. A CloudOps Engineer needs to provide internet access for several Amazon EC2 instances in a private subnet. The Administrator has deployed a NAT instance using an `amzn-ami-vpc-nat` AMI, updated security groups, and configured the appropriate route within the route table. The instances still cannot access the internet. What should be done to resolve the issue?

- A. Disable source/destination checks on the NAT instance.
- B. Start/stop the NAT instance so it is launched on a different host.
- C. Configure a VPN and route via the company proxy server.
- D. Delete the NAT instance and replace it with an internet gateway.

Answer A

52. A company plans to use Amazon Route 53 to enable high availability for a website running on-premises. The website consists of an active and passive server. Route 53 must be configured to route traffic to the primary active server if the associated health returns a 2xx status code. All other traffic should be directed to the secondary passive server. A CloudOps Engineer needs to configure the record type and health check. Which options should the Administrator choose?

- A. An alias record with evaluate health set to yes and associated with a Route 53 HTTP health check.
- B. An A record for each server with an Amazon Route 53 TCP health check.
- C. An A record for each server with an Amazon Route 53 HTTP health check.
- D. An alias record with evaluate health set to yes and associated with a Route 53 TCP health check.

Answer C

53. A team of Analysts require a specialized Amazon EC2 configuration. The team needs to be able to launch and terminate instances across the company's AWS accounts but do not wish to configure EC2 settings on their own. The specialized EC2 configuration contains licensed software and must be available for use only by the Analysts. Which solution should a CloudOps Engineer use to allow the Analysts to deploy their workloads with MINIMAL effort?

- A. Create an AWS CloudFormation template and use it to create a portfolio in AWS Service Catalog. Grant the Analysts permissions to launch products from the portfolio.
- B. Create an AWS Elastic Beanstalk environment. Share the environment across accounts and use IAM policies to enable access for the team of Analysts.
- C. Create an AWS CloudFormation template and configure launch permissions on the AMI used by the template to add the authorized accounts. Share the template in an Amazon S3 bucket.
- D. Create an Amazon Machine Image (AMI) encrypted with an AWS KMS key. Share the encrypted AMI with authorized accounts. Allow the Analysts access to use the KMS key.

Answer A

54. A web application runs on two Amazon EC2 instances behind an Application Load Balancer (ALB). There have been reports from users of poor performance and HTTP 503 and 504 errors. A CloudOps Engineer has reviewed Amazon CloudWatch metrics and discovered that the CPU utilization on the instances is extremely high. Which action should the Administrator take to resolve these issues?

- A. Enable sticky sessions on the Application Load Balancer.
- B. Enable cross-zone load balancing on the Application Load Balancer.
- C. Place the EC2 instances into an Amazon EC2 Auto Scaling group.
- D. Configure the load balancer to use a TCP listener instead of HTTPS.

Answer C

55. An organization deploys a critical application on Amazon EC2 instances within an Auto Scaling group. The application encounters extended scaling times due to lengthy startup scripts. A CloudOps Engineer has been tasked with finding a solution to shorten scale-out durations without exceeding the Auto Scaling group's capacity. Which approach meets these requirements?

- A. Integrate a warm pool into the Auto Scaling group configuration.
- B. Add more instances to the Auto Scaling group to pre-emptively accommodate the increased traffic.
- C. Switch to a larger EC2 instance type in the Auto Scaling group to improve instance performance.
- D. Implement scheduled scaling to increase the instance count during known peak usage periods.

Answer A

56. A CloudOps Engineer manages an application that uses an Application Load Balancer (ALB) in front of six Amazon EC2 instances in a single security group. The Administrator notices that the CloudWatch metric for HealthyHostCount has dropped from 6 to 2. What is MOST likely causing this issue?

- A. The ALB health checks have failed, and the ALB has taken EC2 instances out of service.
- B. The Amazon Route 53 health checks have failed, and the ALB has taken EC2 instances out of service.
- C. The security groups of the instances are not allowing the ALB health checks to succeed.
- D. The route tables are not updated to allow traffic to flow between the ALB and the EC2 instances.

Answer A

57. A critical Amazon EC2 instance has stopped responding. The CloudOps Engineer checked the EC2 management console and noticed that the system status checks are impaired. What first step should the CloudOps Engineer take to resolve this issue?

- A. Take a snapshot, create an AMI, and launch a new instance from the AMI.
- B. Reboot the EC2 instance so it can be launched on a new host.
- C. Terminate the EC2 instance and launch a replacement in another AZ.
- D. Stop and then start the EC2 instance so that it can be launched on a new host.

Answer D

58. An enterprise has mandated the logging of all operations within its AWS account via AWS CloudTrail. In addition, the IT operations team needs to be alerted when any changes or deletions occur in the CloudTrail log files. What is the appropriate approach for the IT operations team to achieve these requirements?

- A. Enable log file integrity validation and use the AWS CloudTrail Processing Library to validate the log files.
- B. Use AWS CloudTrail Insights for monitoring the changes in the log files.
- C. Enable log file integrity validation and use AWS CLI commands to verify the log files.
- D. Use Amazon CloudWatch Events to monitor the modifications in the log file.

Answer C

59. A web application for a popular online store is using Amazon ElastiCache with Memcached for frequently used data. Performance has been poor for a couple of weeks and complaints about the user experience have been reported. The metric data for the Amazon EC2 instances and the Amazon RDS instance appear normal, but the eviction count metrics are high. What can a CloudOps Engineer do to address this issue and resolve the performance issues?

- A. Scale the RDS instance by changing instance types.
- B. Scale the Memcached cluster by adding read replicas.
- C. Scale the Memcached cluster by increasing CPU capacity.
- D. Scale the Memcached cluster by adding additional nodes.

Answer D

60. A company is using an AWS Storage Gateway volume gateway configuration running on a virtual machine. Usage of the Storage Gateway has recently increased, and users have reported that performance of the iSCSI drives has degraded. A CloudOps Engineer checked the Amazon CloudWatch metrics and noticed the CacheHitPercent metric is below 55% and the CachePercentUsed metric is above 95%. What steps are MOST likely to resolve the performance issues?

- A. Create a recovery snapshot from the volume, then deploy and activate a new volume gateway from the snapshot.
- B. Create a larger disk for the cached volume. In the AWS Management Console, edit the local disks, then select the new disk as the cached volume.
- C. Optimize the iSCSI settings on the Storage Gateway's iSCSI initiator to achieve higher I/O performance.
- D. Ensure that the physical disks for the Storage Gateway are in a RAID 1 configuration to allow higher throughput.

Answer B

61. A company needs a way to share Amazon RDS database snapshots of an encrypted database across different AWS accounts they own. The database must be encrypted at rest in the destination accounts. How can the Administrator share the snapshots?

- A. Create a new unencrypted RDS instance from the encrypted snapshot, copy the database contents to a file and then share the file with the AWS accounts.
- B. Take an encrypted snapshot of the RDS database. Share the snapshot with the AWS accounts and then launch an encrypted database from the snapshot.
- C. Update the key policy to grant access to the target accounts. Copy the snapshot using the CMK and then share the snapshot with the target accounts.
- D. Take an unencrypted snapshot of the RDS database. Share the snapshot with the AWS accounts and then launch an encrypted database from the snapshot.

Answer C

62. A company uses a NAT instance to enable Amazon EC2 instances in private subnets to download software and updates from the internet. Latency on the NAT instance has increased as more EC2 instances have been deployed into the private subnets. A CloudOps Engineer needs to reduce latency cost-efficiently and ensure the solution can scale with increasing demand. Which action should be taken to accomplish this?

- A. Change the instance type to use a larger instance type.
- B. Replace the NAT instance with a virtual private gateway.
- C. Replace the NAT instance with a NAT gateway.
- D. Add Auto Scaling for the NAT instance.

Answer C

63. A corporation is operating a hybrid environment consisting of a local data center and an AWS Cloud setup. There are two software applications, one running on the company's on-premises data center named `server1.localcorp.net`, and another one operating on an Amazon EC2 instance known as `server1.awscorp.net`. A stable AWS Site-to-Site VPN connection links the on-premises network to AWS. When the on-premises application attempts to connect to the EC2-hosted application, DNS resolution doesn't seem to work. To allow DNS resolution between resources in the local data center and AWS, a CloudOps Engineer needs to implement a solution. What would allow the on-premises application to successfully resolve the hostname of the EC2 instance?

- A. Implement an outbound Amazon Route 53 Resolver rule in the VPC where the EC2 instance is hosted. Configure the on-premises DNS resolver to forward `localcorp.net` DNS queries to the inbound resolver endpoint.
- B. Enable AWS Direct Connect between the on-premises network and the AWS VPC. Create a conditional forwarding rule in the on-premises DNS resolver to forward all queries to Amazon Route 53.
- C. Use an AWS Transit Gateway to allow DNS resolution between the on-premises network and AWS VPC. Create Amazon Route 53 health check DNS resolution is operating correctly.
- D. Implement an inbound Amazon Route 53 Resolver rule in the VPC where the EC2 instance is running. Configure the on-premises DNS resolver to forward `awscorp.net` DNS queries to the inbound resolver endpoint.

Answer D

64. A CloudOps Engineer has configured a static website on Amazon S3. The bucket name is "mystaticsite". The Administrator plans to use Amazon Route 53 to route traffic to the website and has configured an A record for `www.mystaticsite.com`. However, users are unable to connect to `www.mystaticsite.com` using their browsers. Which of the following is the cause of this issue?

- A. An Amazon CloudFront distribution must be configured for the S3 bucket.
- B. The S3 bucket name must match the record set name in Route 53.
- C. The Route 53 record set requires a certificate to be configured.
- D. The Route 53 record set must be in the same region as the S3 bucket

Answer B

65. (Select Two) An AWS CloudOps Engineer has been tasked with the configuration of a high-performance computing (HPC) application, which is intended to operate on a resilient tier of Amazon EC2 instances. To ensure the smooth running of the application, it is critical that the latency between nodes is kept to a bare minimum. How should the CloudOps Engineer proceed to achieve this?

- A. Launch the EC2 instances into a spread placement group.
- B. Deploy the EC2 instances across multiple subnets to distribute the load.
- C. Launch the EC2 instances in an Auto Scaling group within a single subnet.
- D. Launch the EC2 instances in an Auto Scaling group across Availability Zones.
- E. Configure the EC2 instances in a cluster placement group.

Answer C and E

66. A company has deployed an application on Amazon EC2 instances behind an Application Load Balancer (ALB). There have been reports of errors from users who have attempted to use the application. The CloudOps Engineer noticed an increase in the HTTPCode_ELB_5XX_Count Amazon CloudWatch metric for the load balancer. What is a possible cause for this increase?

- A. The ALB security group does not allow inbound traffic from the users.
- B. The ALB is associated with private subnets within the VPC.
- C. The service limits for ALBs in the VPC has been exceeded.
- D. The target group of the ALB does not contain healthy EC2 instance

Answer D

67. A legacy application is deployed on an Amazon EC2 m1.large instance and the CPU utilization is over 90% resulting in high latency. The application can only be scaled vertically. How can a CloudOps Engineer resolve the performance issues?

- A. Upgrade to a compute-optimized instance.
- B. Change the Amazon EBS volume to Provisioned IOPS.
- C. Add a read replica to offload queries from the instance.
- D. Add additional m1.large instances to the application.

Answer A

68. A root account owner has given full access of his S3 bucket to one of the IAM users using the bucket ACL. When the IAM user logs in to the S3 console, which actions can he perform?

- A. He can just view the content of the bucket.
- B. He can do all the operations on the bucket.
- C. It is not possible to give access to an IAM user using ACL.
- D. The IAM user can perform all operations on the bucket using only API/SDK.

Answer C

69. A company wants to build a solution for its business-critical Amazon RDS for MySQL database. The database requires high availability across different geographic locations. A CloudOps Engineer must build a solution to handle a Disaster Recovery (DR) scenario with the lowest Recovery Time Objective (RTO) and Recovery Point Objective (RPO). Which solution meets these requirements?

- A. Create automated snapshots of the database on a schedule. Copy the snapshots to the DR Region.
- B. Create a Cross-Region read replica for the database.
- C. Create a Multi-AZ read replica for the database.
- D. Schedule AWS Lambda functions to create snapshots of the source database and to copy the snapshots to a DR Region

Answer B

70. A company runs a stateless application that is hosted on an Amazon EC2 instance. Users are reporting performance issues. A CloudOps Engineer reviews the Amazon CloudWatch metrics for the application and notices that the instance's CPU utilization frequently reaches 90% during business hours. What is the MOST operationally efficient solution that will improve the application's responsiveness?

- A. Configure CloudWatch logging on the EC2 instance. Configure a CloudWatch alarm for CPU utilization to alert the CloudOps Engineer when CPU utilization goes above 90%.
- B. Configure an AWS Client VPN connection to allow the application users to connect directly to the EC2 instance private IP address to reduce latency.
- C. Create an Auto Scaling group, and assign it to an Application Load Balancer. Configure a target tracking scaling policy that is based on the average CPU utilization of the Auto Scaling group.
- D. Create a CloudWatch alarm that activates when the EC2 instance's CPU utilization goes above 80%. Configure the alarm to invoke an AWS Lambda function that vertically scales the instance

Answer C

71. A CloudOps Engineer must create a solution that automatically shuts down any Amazon EC2 instances that have less than 10% average CPU utilization for 60 minutes or more. Which solution will meet this requirement In the MOST operationally efficient manner?

- A. Implement a cron job on each EC2 instance to run once every 60 minutes and calculate the current CPU utilization. Initiate an instance shutdown If CPU utilization is less than 10%.
- B. Implement an Amazon CloudWatch alarm for each EC2 instance to monitor average CPU utilization. Set the period at 1 hour, and set the threshold at 10%. Configure an EC2 action on the alarm to stop the instance.
- C. Install the unified Amazon CloudWatch agent on each EC2 instance, and enable the Basic level predefined metric set. Log CPU utilization every 60 minutes, and initiate an instance shutdown if CPU utilization is less than 10%.
- D. Use AWS Systems Manager Run Command to get CPU utilization from each EC2 instance every 60 minutes. Initiate an instance shutdown if CPU utilization is less than 10%.

Answer B

72. A CloudOps Engineer is setting up an automated process to recover an Amazon EC2 instance in the event of an underlying hardware failure. The recovered instance must have the same private IP address and the same Elastic IP address that the original instance had. The SysOps team must receive an email notification when the recovery process is initiated. Which solution will meet these requirements?

- A. Create an Amazon CloudWatch alarm for the EC2 instance, and specify the `StatusCheckFailedInstance` metric. Add an EC2 action to the alarm to recover the instance. Add an alarm notification to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the SysOps team email address to the SNS topic.
- B. Create an Amazon CloudWatch alarm for the EC2 instance, and specify the `StatusCheckFailed_System` metric. Add an EC2 action to the alarm to recover the instance. Add an alarm notification to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the SysOps team email address to the SNS topic.
- C. Create an Auto Scaling group across three different subnets in the same Availability Zone with a minimum, maximum, and desired size of 1. Configure the Auto Scaling group to use a launch template that specifies the private IP address and the Elastic IP address. Add an activity notification for the Auto Scaling group to send an email message to the SysOps team through Amazon Simple Email Service (Amazon SES).
- D. Create an Auto Scaling group across three Availability Zones with a minimum, maximum, and desired size of 1. Configure the Auto Scaling group to use a launch template that specifies the private IP address and the Elastic IP address. Add an activity notification for the Auto Scaling group to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the SysOps team email address to the SNS topic.

Answer B

73. A CloudOps Engineer is troubleshooting connection timeouts to an Amazon EC2 instance that has a public IP address. The instance has a private IP address of 172.31.16.139. When the CloudOps Engineer tries to ping the instance's public IP address from the remote IP address 203.0.113.12, the response is request timed out. The flow logs contain the following information: What is one cause of the problem?

- A. Inbound security group deny rule.
- B. Outbound security group deny rule.
- C. Network ACL inbound rules.
- D. Network ACL outbound rules.

Answer D

74. A company has a high-performance Windows workload. The workload requires a storage volume that provides consistent performance of 10,000 IOPS. The company does not want to pay for additional unneeded capacity to achieve this performance. Which solution will meet these requirements with the LEAST cost?

- A. Use a Provisioned IOPS SSD (io1) Amazon Elastic Block Store (Amazon EBS) volume that is configured with 10,000 provisioned IOPS.
- B. Use a General Purpose SSD (gp3) Amazon Elastic Block Store (Amazon EBS) volume that is configured with 10,000 provisioned IOPS.
- C. Use an Amazon Elastic File System (Amazon EFS) file system in Max I/O mode.
- D. Use an Amazon FSx for Windows File Server file system that is configured with 10,000 IOPS.

Answer B

75. A company runs thousands of Amazon EC2 instances that are based on the Amazon Linux 2 Amazon Machine Image (AMI). A CloudOps Engineer must implement a solution to record commands and output from any user that needs an interactive session on one of the EC2 instances. The solution must log the data to a durable storage location. The solution also must provide automated notifications and alarms that are based on the log data. Which solution will meet these requirements with the MOST operational efficiency?

- A. Configure command session logging on each EC2 instance. Configure the unified Amazon CloudWatch agent to send session logs to Amazon CloudWatch Logs. Set up query filters and alerts by using Amazon Athena.
- B. Require all users to use a central bastion host when they need command line access to an EC2 instance. Configure the unified Amazon CloudWatch agent on the bastion host to send session logs to Amazon CloudWatch Logs. Set up a metric filter and a metric alarm for relevant security findings in CloudWatch Logs.
- C. Require all users to use AWS Systems Manager Session Manager when they need command line access to an EC2 instance. Configure Session Manager to stream session logs to Amazon CloudWatch Logs. Set up a metric filter and a metric alarm for relevant security findings in CloudWatch Logs.
- D. Configure command session logging on each EC2 instance. Require all users to use AWS Systems Manager Run Command documents when they need command line access to an EC2 instance. Configure the unified Amazon CloudWatch agent to send session logs to Amazon CloudWatch Logs. Set up CloudWatch alarms that are based on Amazon Athena query results.

Answer C

76. A company uses AWS CloudFormation to deploy its infrastructure. The company recently retired an application. A cloud operations engineer initiates CloudFormation stack deletion, and the stack gets stuck in DELETE_FAILED status. A CloudOps Engineer discovers that the stack had deployed a security group. The security group is referenced by other security groups in the environment. The CloudOps Engineer needs to delete the stack without affecting other applications. Which solution will meet these requirements in the MOST operationally efficient manner?

- A. Create a new security group that has a different name. Apply identical rules to the new security group. Replace all other security groups that reference the new security group. Delete the stack.
- B. Create a CloudFormation change set to delete the security group. Deploy the change set.
- C. Delete the stack again. Specify that the security group be retained.
- D. Perform CloudFormation drift detection. Delete the stack

Answer B

77. A company has scientists who upload large data objects to an Amazon S3 bucket. The scientists upload the objects as multipart uploads. The multipart uploads often fail because of poor end-client connectivity. The company wants to optimize storage costs that are associated with the data. A CloudOps Engineer must implement a solution that presents metrics for incomplete uploads. The solution also must automatically delete any incomplete uploads after 7 days. Which solution will meet these requirements?

- A. Review the Incomplete Multipart Upload Bytes metric in the S3 Storage Lens dashboard. Create a S3 Lifecycle policy to automatically delete any incomplete multipart uploads after 7 days.
- B. Implement S3 Intelligent-Tiering to move data into lower-cost storage classes after 7 days. Create a S3 Storage Lens policy to automatically delete any incomplete multipart uploads after 7 days.
- C. Access the S3 console. Review the Metrics tab to check the storage that incomplete multipart uploads are consuming. Create an AWS Lambda function to delete any incomplete multipart uploads after 7 days.
- D. Use the S3 analytics storage class analysis tool to identify and measure incomplete multipart uploads. Configure a S3 bucket policy to enforce restrictions on multipart uploads to delete incomplete multipart uploads after 7 days.

Answer A

78. A company is uploading important files as objects to Amazon S3. The company needs to be informed if an object is corrupted during the upload. What should a CloudOps Engineer do to meet this requirement?

- A. Pass the Content-Disposition value as a request body during the object upload.
- B. Pass the Content-MD5 value as a request header during the object upload.
- C. Pass x-amz-object-lock-mode as a request header during the object upload.
- D. Pass x-amz-server-side-encryption-customer-algorithm as a request body during the object upload

Answer B

79. A company is supposed to receive a data file every hour in an Amazon S3 bucket. A S3 event notification invokes an AWS Lambda function each time a file arrives. The function processes the data for use by an application. The application team notices that sometimes the file does not arrive. The application team wants to receive a notification whenever the file does not arrive. What is the MOST operationally efficient solution that meets these requirements?

- A. Add a S3 Lifecycle rule on the S3 bucket with a scope that is limited to objects that were created in the last hour. Configure another S3 event notification to be invoked by the lifecycle transition when the number of objects transitioned is zero. Publish a message to an Amazon Simple Notification Service (Amazon SNS) topic to notify the application team.
- B. Configure another S3 event notification to invoke a Lambda function that posts a message to an Amazon Simple Queue Service (Amazon SQS) queue. Create an Amazon CloudWatch alarm to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic to notify the application team when the ApproximateAgeOfOldestMessage metric of the queue is greater than 1 hour.
- C. Create an Amazon CloudWatch alarm to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic to alert the application team when the Invocations metric of the Lambda function is zero for an hour. Configure the alarm to treat missing data as breaching.
- D. Create a new Lambda function to get the timestamp of the newest file in the S3 bucket. If the timestamp is more than 1 hour ago, publish a message to an Amazon Simple Notification Service (Amazon SNS) topic to notify the application team. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to invoke the new function hourly.

Answer C

80. A CloudOps Engineer is provisioning an Amazon Elastic File System (Amazon EFS) file system to provide shared storage across multiple Amazon EC2 instances. The instances all exist in the same VPC across multiple Availability Zones. There are two instances in each Availability Zone. The CloudOps Engineer must make the file system accessible to each instance with the lowest possible latency. Which solution will meet these requirements?

- A. Create a mount target for the EFS file system in the VPC. Use the mount target to mount the file system on each of the instances.
- B. Create a mount target for the EFS file system in one Availability Zone of the VPC. Use the mount target to mount the file system on the instances in that Availability Zone. Share the directory with the other instances.
- C. Create a mount target for each instance. Use each mount target to mount the EFS file system on each respective instance.
- D. Create a mount target in each Availability Zone of the VPC. Use the mount target to mount the EFS file system on the instances in the respective Availability Zone.

Answer D

81. A CloudOps Engineer creates an AWS CloudFormation template to define an application stack that can be deployed in multiple AWS Regions. The CloudOps Engineer also creates an Amazon CloudWatch dashboard by using the AWS Management Console. Each deployment of the application requires its own CloudWatch dashboard. How can the CloudOps Engineer automate the creation of the CloudWatch dashboard each time the application is deployed?

- A. Create a script by using the AWS CLI to run the `aws cloudformation put-dashboard` command with the name of the dashboard. Run the command each time a new CloudFormation stack is created.
- B. Export the existing CloudWatch dashboard as JSON. Update the CloudFormation template to define an `AWS::CloudWatch::Dashboard` resource. Include the exported JSON in the resource's `DashboardBody` property.
- C. Update the CloudFormation template to define an `AWS::CloudWatch::Dashboard` resource. Use the `Intrinsic Ref` function to reference the ID of the existing CloudWatch dashboard.
- D. Update the CloudFormation template to define an `AWS::CloudWatch::Dashboard` resource. Specify the name of the existing dashboard in the `DashboardName` property.

Answer B